Differentiation

Limiting values of functions. Generalising the concept of sequence limit to function values.

We say $\lim_{x \to a} f(x) = l$ makes not explicit reference to the value $f(a)$

to mean $f(x)$ converge to l as x converges to a. (informal)

For the example shown in fig 4.1.1. there would not exist a limiting value l for this f as $x \to 1$.

Can prove similar properties for function limit values as for sequences.

This theorem can be proved by using the more detailed formal definition of limiting value.

Formal def

$$\lim_{x \to a} f(x) = l \qquad \delta \in \mathbb{R}$$

$\Leftrightarrow$ for all $\varepsilon > 0$ there exists a $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - l| < \varepsilon$

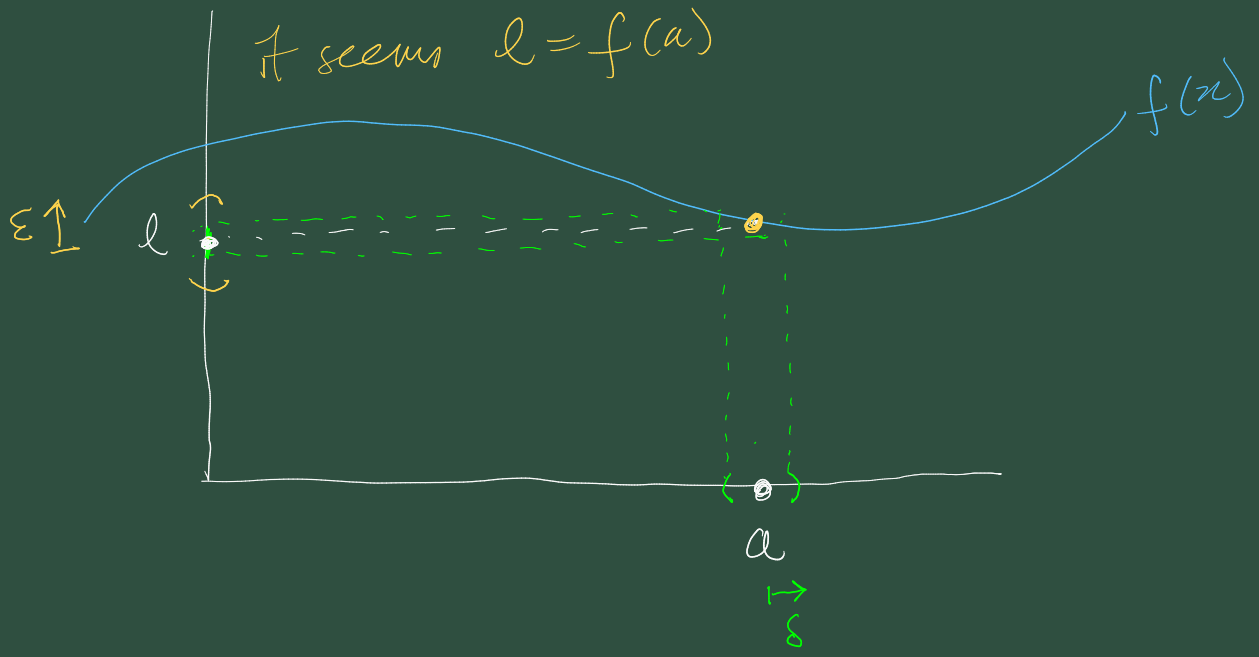
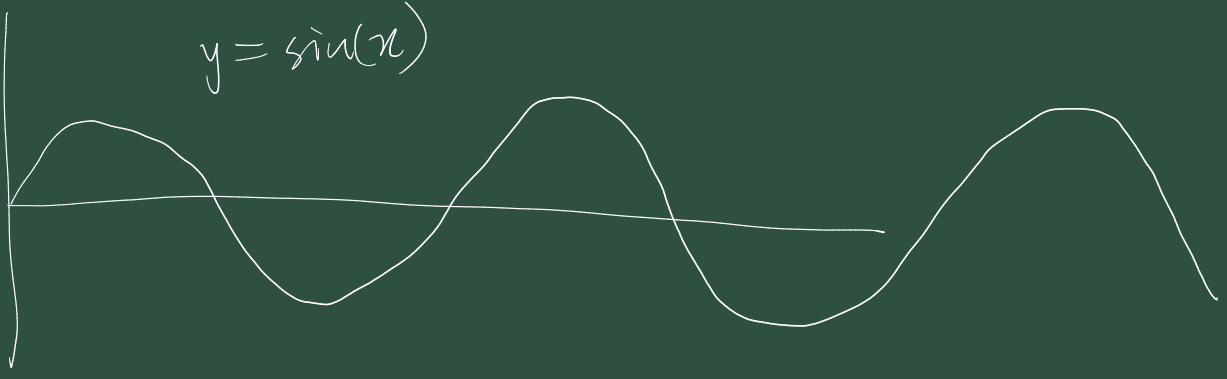


Fig 4.1.1 also shows a function not continuous at 1.



In figure 4.2.2. we see $y = \sin(1/x)$ a function that is continuous everywhere except at $x = 0$. Indeed there is no limiting value for $\lim_{x \to 0} \sin\left(\frac{1}{x}\right)$!

Formal def. of continuity is the formal def. for limiting value with $l = \lim_{x \to a} f(x) = f(a)$

4.3 The derivative.

Consider the bare concept of change in value i.e.

$$f(x) - f(a)$$

can improve this by making this change relative to how much x, a have changed

$$\frac{f(x) - f(a)}{x - a}$$

the "relative change in value of f, between x and a" = gradient of the line through $(a, f(a))$ and $(x, f(x))$

To capture ~~this~~ the nature of f at a we consider

$$f'(a) = \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \right)$$ "f prime"

the "derivative of f at a".

Also written as $\left.\frac{df}{dx}\right|_{x=a} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$

It is the gradient of the "tangent line to f

at a".

Some examples For easy enough functions we can evaluate $f'(a)$ directly from the limit definition.

For $f(x) = kx$, $k \in \mathbb{R}$.

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{x \to a} \frac{kx - ka}{x - a}$$

$$= \lim_{x \to a} \frac{k(x-a)}{(x-a)}$$

$$= \lim_{x \to a} k$$

$= k$, since k is a constant as $x \to a$.

For $g(x) = kx^2$

$$g'(a) = \lim_{x \to a} \frac{kx^2 - ka^2}{x - a}$$

$$= \lim_{x \to a} \frac{k(x-a)(x+a)}{x-a}$$

$$= \lim_{x \to a} k(x+a)$$

$= k(a+a)$, by the algebra of limits theorem for function values.

$= 2ka$,

This was the "first" derivative of a function. We can iterate this operation to obtain the higher-order derivatives.

$$\frac{d^2f}{dx^2} = \frac{d}{dx}\left(\frac{df}{dx}\right), \quad \frac{d^3f}{dx^3} = \frac{d}{dx}\left(\frac{d^2f}{dx^2}\right)$$

and so on.

In the prime notation we can write.

$$f''(a) = \left.\frac{d^2f}{dx^2}\right|_{x=a}.$$

$$\vdots$$

$$f^{(n)}(a) = \left.\frac{d^nf}{dx^n}\right|_{x=a}$$

(Continue at 12:05)

Properties of the derivative

How does the derivative of a 'complex' function depend on the derivatives of the functions it is built from.

Theorem 4.4.1

Given functions g, h and $\alpha, \beta \in \mathbb{R}$.

Define $f(x) = \alpha g + \beta h$.

then $f'(a) = \alpha g'(a) + \beta h'(a)$

Proof. Done using the algebra of limits theorem for limiting values.

$$= \ldots\ldots = \underbrace{\lim_{x \to a} \frac{f(x) - f(a)}{x - a}}$$

$$= f'(a).$$

Theorem 4.4.2 Product rule.

Given 2 differentiable functions g, h define.

$f = gh$, ie. $f(x) = g(x)\,h(x)$

WARNING

$f'(a) \neq g'(a)\,h'(a)$

Rather, the product rule is.

$$f'(a) = g'(a)\,h(a) + g(a)\,h'(a)$$

Proof.

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

$$= \lim_{x \to a} \frac{g(x)h(x) - g(a)h(a)}{x - a}.$$

$$= \lim_{x \to a} \frac{g(x)\big(h(x) - h(a)\big) + h(a)\big(g(x) - g(a)\big)}{x - a}$$

$$= \lim_{x \to a} \left(\frac{g(x)\big(h(x) - h(a)\big)}{x - a} \right) + \lim_{x \to a} \left(\frac{h(a)\big(g(x) - g(a)\big)}{x - a} \right)$$

, by algebra of limits.

$$= \lim_{x \to a} \big(g(x)\big) \lim_{x \to a} \left(\frac{h(x) - h(a)}{x - a} \right)$$

, by algebra of limits theorem $+ \; h(a) \lim_{x \to a} \left(\frac{g(x) - g(a)}{x - a} \right)$

$$= g(a)h'(a) + h(a)g'(a), \text{ as required.}$$

The other main ways of forming new functions are composition and quotients.

Theorem 4.4.6. Composition / chain rule.

Given functions f, g where g is differentiable at a and f is differentiable at $g(a)$, then consider their composition

$$y = f \circ g$$

i.e. $y(x) = f(g(x))$

then

$$y'(a) = f'(g(a))\, g'(a)$$

Prove it soon. (next week).

But let's use it to establish the quotient rule.

Theorem 4.4.8 Define y as the quotient of functions u, v

i.e. $y = \dfrac{u}{v}$

then $y'(a) = \dfrac{v(a)u'(a) - u(a)v'(a)}{v(a)^2}$

Let's prove this using the product rule and chain rule. We can rewrite the quotient y as a product.

$$y = \frac{u}{v} = u \cdot \left(\frac{1}{v}\right)$$

$$= u \cdot (r \circ v)$$

where r is the reciprocal function.

ie. $r(x) = \frac{1}{x}$.

Lemma $\quad r'(a) = \frac{-1}{a^2}$

Proof:

$$r'(a) = \lim_{x \to a} \frac{r(x) - r(a)}{x - a}$$

$$= \lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$$

$$= \lim_{x \to a} \left(\frac{\frac{a - x}{xa}}{x - a} \right)$$

$$= \lim_{x \to a} \left(\frac{-(x - a)}{xa\,(x - a)} \right)$$

$a \neq 0$.

$$= -\lim_{n \to a} \left(\frac{1}{na} \right)$$

$$= -\frac{1}{a^2}$$

Recall $y = \frac{u}{v} = u \cdot (r \circ v)$

$$y'(a) = u'(a)\,(r \circ v)(a) + u(a)\,(r \circ v)'(a)$$

, by the product rule.

$$= u'(a)\,\frac{1}{v(a)} + u(a)\,r'(v(a))\,v'(a)$$

, by the def of r and the chain rule.

$$= \frac{u'(a)}{v(a)} + u(a)\left(-\frac{1}{v(a)^2}\right) v'(a)$$

$$= \frac{u'(a)\,v(a)}{v^2(a)} - \frac{u(a)\,v'(a)}{v(a)^2}$$

$$= \frac{u'(a)\,v(a) - u(a)\,v'(a)}{v(a)^2}$$, as required.

Next week Proof of chain rule after considering "linear approximation with error".